Independent Verification of a Four-Storey Braced Steel Frame

Revit to Oasys GSA: what the handoff breaks, and what the model reveals that hand calculation cannot

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Self-directed project. Model, analysis and code at

github.com/Abgar77/bim-to-analysis

1 Introduction

This report records an independent verification of a four-storey braced steel frame that was originally designed by hand to EN 1993-1-1 as coursework. The frame was rebuilt as a Revit structural model, exported to Oasys GSA, and the finite element results compared, member by member, against the original calculations.

The purpose was not to establish whether GSA computes correctly. With proven software the analysis will be correct; the exercise is to check the engineer's input, and to confirm that an appropriate structural model has been used. The question the work actually set out to answer was narrower and more useful: what does it take to turn a BIM model into a trustworthy analysis model, and what does the handoff silently break?

Three layers of answer emerged, and the report is organised around them. The first is data: section properties did not survive the export, and the geometry arrived in thirty-four disconnected pieces, both without warning. The second is idealisation: the design's own assumption of nominally pinned connections cannot be solved in a finite element model at all, and its tension-only bracing removes the superposition that combination cases depend on. The third is result: once the model solves, it measures four quantities the hand method is structurally incapable of producing — column moments, truss chord secondary bending, deflection accumulated down a transfer load path, and the elastic critical load factor.

The last of these is the substantive finding. The design used first-order analysis throughout with buckling length equal to storey height. The model returns an elastic critical load factor of 4.58, which places the frame in the band where EN 1993-1-1 §5.2.1 requires second-order effects to be accounted for.

2 The structure

The building is a four-storey braced steel office, 24.5 m by 32.5 m on plan and 22 m tall, in S355 steel with a 130 mm composite slab on profiled metal decking. Gridlines 1 to 6 run at 0, 5000, 10000, 14500, 19500 and 24500 mm; rows A to F at 6500 mm centres. Levels are at 0, 7000, 12000, 17000 and 22000 mm.

The defining feature is a transfer truss at Level 1 on each of rows B, C, D and E. Each truss spans the full 24.5 m at 2.45 m depth in a modified Warren configuration, and carries the internal C2 columns so that the ground floor is free of internal columns. Lateral stability is provided by X-bracing on the perimeter in both directions, and all connections are nominally pinned — simple construction in the UK sense.

Member	Section	Design action	Utilisation
B1 secondary	UKB 356×127×33	M = 173.70 kNm	0.901
B2 secondary	UKB 457×152×52	M = 333.41 kNm	0.854
B3 secondary	UKB 356×171×51	M = 317.46 kNm	0.998
B4 primary	UKB 610×229×113	M = 1058.98 kNm	0.909
C1 column	UKC 356×368×177	N = 3941.51 kN	0.877
C2 column	UKC 254×254×73	N = 1603.60 kN	0.915
Truss chord	UKC 356×406×235	N = 9383 kN	0.926
Truss diagonal	UKC 203×203×86	N = 2837.62 kN	0.982
Brace BR-X	UKA 70×50×6	N = 232.74 kN	0.955
Brace BR-Y	UKA 60×30×5	N = 149.88 kN	0.993

Table 1 Governing members from the original design, as frozen in the baseline register.

Several members were sized close to capacity. B3 passed by 0.62 kNm and BR-Y by less than one per cent, which made them useful indicators of whether any modelling difference was material.

3 Method

3.1 Baseline freeze

Before any model was built, every member's design forces, capacities, utilisations and governing check were extracted from the original report into a single register of twenty-six member types, committed to version control and not edited afterwards. Predicted discrepancies were written down at the same time, with the arithmetic behind each. Comparison against a record fixed in advance is what separates verification from rationalisation after the fact.

3.2 Software chain

The physical model was built in Revit 2023 — the last release supported by the GsRevit link — and exported to Oasys GSA 10.2 through that plugin. Analysis, model stability diagnostics, modal buckling and result extraction were carried out in GSA. Comparison tables and derived calculations were produced in Python. All models, exports and scripts are in the repository.

3.3 Basis of comparison

Acceptance bands were declared in advance: under 2 per cent counted as agreement, 2 to 10 per cent required an explanation, and anything beyond 10 per cent was treated as an error until proven otherwise. Every difference was then classified as one of three things — a model error, an idealisation difference where both positions are defensible, or quantified conservatism in the hand method.

4 The BIM to FE transfer

4.1 Section properties did not transfer

All twelve sections arrived in GSA as explicit (EXP) profiles with area, second moment of area, torsion constant and section modulus all set to zero. The section names were correct and the model looked right on screen; it simply had no stiffness anywhere. The cause was that no section mapping database was built during export, so the plugin had nothing to map Revit families onto GSA catalogue entries.

The fix was to enter catalogue references by hand for each property. Two observations are worth recording. First, one angle resolved to UA 75×50×6 rather than the designed 70×50×6, a substitution that would have propagated silently into the brace check. Second, two section references were flagged by GSA as superseded, which is a reminder that catalogue vintage is itself an input.

4.2 Thirty-four disconnected structures

On import the model comprised 440 members and 356 nodes, resolving into thirty-four separate structures rather than one. The cause was not random tolerance error but three distinct and traceable coordinate offsets.

Offset	Attributable to
50 mm	Differing beam depths under top-of-steel justification
130 mm	Slab thickness
180 mm	Half a chord depth under centre justification

Table 2 Coordinate offsets found at import, each traceable to a Revit justification convention.

The underlying incompatibility is that Revit positions analytical lines from section geometry, while GSA requires members to share nodes. The two conventions do not reconcile, and nothing in the export warns that they have failed to. Resolution required staged node merging at 10, 60 and 140 mm together with direct coordinate edits, reducing 356 nodes to 220 and thirty-four structures to one.

4.3 The diaphragm tool produced corner links

Applying GSA's rigid diaphragm to the 2D floor members generated four-node constraints at the slab corners only, rather than tying each floor. The floors were therefore switched to load panel analytical type so that they would distribute load without pretending to provide lateral stiffness. This is recorded here because the same behaviour recurred during the stability work, where it mattered a great deal — see §9.2.

5 Idealisation and solvability

5.1 The fully pinned model cannot be solved

The design assumes nominally pinned connections throughout. Modelling that directly — major-axis, minor-axis and torsional rotations released at both ends of every beam, truss web members as struts, braces as ties — produced four zero eigenvalues and a stiffness matrix condition number of 7.34 × 10^39. The static solve terminated without producing results.

GSA's model stability analysis identified the cause precisely. Eigenvalues one to four lay between 10^-9 and 10^-12 while mode five was 25,210, a gap of thirteen orders of magnitude indicating four rigid-body mechanisms. The high-stiffness modes were all within 14 per cent of one another, which ruled out the other common cause of ill-conditioning, a disproportionately stiff short element.

The mechanism is straightforward once identified. A node at which every connection is pinned has no rotational stiffness at all. At the truss panel points the weak direction was rotation about the chord's own longitudinal axis, where a continuous chord offers only torsional stiffness GJ — two to three orders of magnitude below EI for an open section. Restoring minor-axis continuity on the truss diagonals resolved it: because they are inclined at 63 degrees, their bending stiffness contributes a large component along the chord axis. The condition number fell to 7.34 × 10^5, an improvement of sixteen orders of magnitude, and the model solved.

The general point is worth stating plainly. Hand calculation of simple construction never assembles a global stiffness matrix, so it never encounters this mechanism. The same idealisation built in finite elements will not solve. Reconciling the two methods requires retaining continuity that the hand method assumes away — the two approaches are not merely different in precision, they rest on assumptions that are not mutually compatible.

5.2 Release scheme adopted

Members	Type	Releases	Matches the design?
C1, C2 columns	Beam	None	No — design assumes pin-ended
Beams B1–B5, BP	Beam	yy both ends	Yes, for major-axis bending
Braces	Tie	None	Yes — tension only
Truss chords	Beam	None	No — design uses method of joints

Members	Type	Releases	Matches the design?
Truss web	Beam	yy both ends	Partly

Table 3 Final release scheme and its relationship to the design assumptions.

Releasing only major-axis rotation leaves minor-axis bending and torsion continuous at every joint. This is arguably closer to a real fin plate than the design's own idealisation: a fin plate bolts through the beam web only, so the beam genuinely rotates in the vertical plane, but the plate resists torsion and minor-axis bending. Releasing all three rotations models a connection that does not exist.

5.3 Tension-only bracing removes superposition

Modelling the braces as ties reproduces the design assumption exactly — one diagonal carries the load and the other goes slack. It also makes the solution nonlinear, because the active stiffness depends on the load state. GSA warns that results from tie, strut and cable elements should not be combined, and rejected combination cases outright.

Factored load states therefore had to be written directly into analysis case expressions so that the solver resolves the correct tie configuration for each complete load state. Once again, an assumption that simplifies hand calculation complicates the numerical model.

6 Loading and validation

6.1 Load panels carry no mass

GSA's gravity loading acts only on beams, bars, ties and struts, computing self-weight from section area, density and length. Load panels have no mass, so the 130 mm slab was excluded entirely. This amounts to 3.06 kN/m² over 3,185 m², or 9,748 kN — about 38 per cent of the serviceability dead load — and had to be applied manually as a face load. An analysis that missed it would produce results that look entirely plausible and are forty per cent light.

6.2 Equilibrium check on every case

Each load case was analysed individually and its applied load compared against the sum of support reactions before any combination was run.

Case	Load	Applied	Reacted
A1	Steel self weight	2.201 MN	2.201 MN
A2	Slab self weight	9.746 MN	9.746 MN
A3	Superimposed dead	2.707 MN	2.707 MN
A4	Imposed	8.361 MN	8.361 MN
A5	Partitions	2.389 MN	2.389 MN
A6	Snow	557.4 kN	557.4 kN
A7	Wind X	270.6 kN	270.6 kN
A8	Wind Y	204.0 kN	204.0 kN
A9	ULS gravity	35.91 MN	35.91 MN
A12	SLS variable	10.75 MN	10.75 MN
A13	SLS all	25.40 MN	25.40 MN

Table 4 Equilibrium validation. Residuals were below 0.002 per cent throughout.

Total steel weight came to 2.201 MN, approximately 224 tonnes or 70 kg/m², which is plausible for a braced frame carrying a transfer structure.

7 Member verification

7.1 Beam self-weight over-assumed

The project brief specified a self-weight allowance of 200 kg/m for every beam. The sections actually selected range from 33 to 113 kg/m. The resulting phantom load is 1.35 × (200 − w) × 9.81/1000 kN/m, and it is largest in proportional terms for the lightest sections.

Member	Section	Baseline	Model	Change
B1	UKB 356×127×33	173.70 kNm	162.0 kNm	−6.7%
B2	UKB 457×152×52	333.41 kNm	323.1 kNm	−3.1%
B3	UKB 356×171×51	317.46 kNm	307.0 kNm	−3.3%
B4	UKB 610×229×113	1058.98 kNm	1013.0 kNm	−4.3%

Table 5 Major-axis design moments, baseline against model, under the ULS gravity combination.

The B1 case can be derived in full. The phantom factored load is 1.35 × (200 − 33) × 9.81/1000 = 2.212 kN/m. The design uniformly distributed load of 32.89 kN/m therefore becomes 30.68 kN/m, and M = wL²/8 over a 6.5 m span gives 162.02 kNm against the original 173.70 kNm. GSA returned 162.0 kNm. Shears track the same ratio, B1 falling from 106.89 kN to 99.7 kN, also 6.7 per cent.

The reduction scales inversely with section weight exactly as the mechanism predicts, which is what makes this a mechanism rather than a coincidence. The practical consequence is that B3, which passed by 0.62 kNm in the original design, now has approximately 11 kNm of margin and no longer governs.

7.2 Truss self-weight over-assumed by 39 per cent

The original report computes the truss self-weight in its own Table 6.1 as 615.19 kg/m, and then continues the design using the brief's 1000 kg/m assumption. The difference is approximately 92 kN of non-existent load per truss across four trusses.

Quantity	Baseline	Model	Change
Chord, compression	9383 kN	8085 kN	−13.8%
Chord, tension	9383 kN	8074 kN	−14.0%

Table 6 Truss chord axial forces.

At fourteen per cent this is the largest single discrepancy identified, and it is an internal inconsistency in the original document rather than an error of judgement — the correct figure was calculated and then not used.

7.3 Column axial forces

Column axial forces fell by approximately ten per cent, more than the beams, because the columns accumulate the truss self-weight error on top of the beam self-weight errors. C1 fell from 3941.51 kN to 3563 kN and C2 from 1603.60 kN to 1446 kN.

A useful corroboration emerged in the C1 stack. The ground-storey axial force is 3563 kN, dropping to 1025 kN above Level 1; approximately 2,538 kN therefore enters at the truss bearing, confirming that the transfer structure carries what it was designed to carry.

8 Effects the design method cannot capture

The findings in this section are not errors. They are quantities the hand method is incapable of producing, because the assumptions that make hand calculation tractable also remove these effects from consideration.

8.1 Truss chords carry secondary bending

The original design derives truss member forces by the method of joints, and states explicitly that "as the members of the truss are pinned, Lcr = L" and that the truss elements are pin jointed. Under that method the chord moment is zero by construction; it is an axiom of the analysis, not a result of it.

Modelled with continuous chords, the peak major-axis moment in the chord is approximately 432 kNm, which is of the order of 26 per cent of the chord's bending capacity. A common rule of thumb treats secondary moments below five per cent of capacity as confirming a pin-jointed idealisation and above ten per cent as invalidating it. The magnitude depends on the continuity assumption, and a real connection detail would modify it, but the effect is clearly not negligible.

8.2 Columns carry moments the design treats as zero

Both column types are designed as pin-ended and concentrically loaded, with buckling length equal to storey height. The model reports a moment of 307.5 kNm in C1 at the truss bearing joint. Screened as a beam-column on a linear interaction, the utilisation rises from 0.877 on axial force alone to 1.044 if the moment acts about the major axis, or 1.311 about the minor axis. C2 remains below unity at 0.970.

This is a consequence of the continuity retained in the model, and the honest statement is that the pinned idealisation is not conservative at this joint. What follows from it is that the connection detail at the truss bearing matters, and the design does not consider it.

8.3 Deflection accumulates down the transfer load path

This is the finding with the clearest practical consequence. Under variable load alone, the transfer truss deflects 30.30 mm at mid-span. It passes every conventional limit; span over 500 would allow 49.0 mm.

But everything the truss carries inherits that displacement. A B3 secondary beam at Level 3 has 18.17 mm of its own sag on top of 31.08 mm of support settlement, reaching 49.25 mm in absolute terms. The load path was confirmed explicitly — truss, to C2 column, to B4 primary, to B3 secondary — and the truss deflection and the support settlement agree to within 0.78 mm, the balance being B4's own sag and column shortening.

Member	Own sag	Support settlement	Absolute	Member limit
B1, Level 3	16.24 mm	1.67 mm	17.91 mm	32.5 mm
B3, Level 3 mid	18.17 mm	31.08 mm	49.25 mm	32.5 mm
Transfer truss	30.30 mm	—	30.30 mm	49.0 mm

Table 7 Deflection under unfactored variable load. Each member passes its own check; the floor does not.

The design checks each member against its own span assuming rigid supports. That is correct practice and it is all that hand calculation permits. What it cannot do is accumulate deflection down a load path — and in a building with a transfer structure, that accumulation is precisely where the serviceability problem lives. Nearly fifty millimetres of floor movement under live load will crack partitions and be noticeable underfoot, and nothing in the original check would reveal it.

Members' own sag came out 7.7 to 8.3 per cent above the hand calculation. Two candidate causes are shear deformation, which GSA includes in its beam elements and the standard $5wL^4/384EI$ expression does not, and the load panel distribution differing from the assumed tributary strip.

9 Stability and second-order effects

9.1 Elastic critical load factor

Modal buckling analysis was run with thirty modes and a lower cut-off of zero, taking geometric stiffness from each factored load case in turn.

Geometric stiffness basis	Mode 1 factor	Global mode participation
ULS gravity	4.821	0.40
ULS + wind X	4.687	0.73
ULS + wind Y	4.576	0.90

Table 8 Lowest buckling load factors. The governing case is wind Y, at 90 per cent global participation.

The governing value is 4.58, and the 90 per cent global mode participation confirms it is a genuine frame mode rather than a local instability in one member. EN 1993-1-1 §5.2.1 sets the threshold for neglecting second-order effects at 10. At 4.58 the frame sits in the $3 \le \alpha_{cr} < 10$ band, where second-order effects must be accounted for. The original design used first-order analysis throughout, with buckling length equal to storey height.

The amplifier of §5.2.2(6B), $1/(1 - 1/\alpha_{cr})$, is 1.28. Applied to the members that were already close to capacity, BR-Y rises from 0.993 to approximately 1.27, BR-X from 0.955 to 1.22, and C1 as a beam-column from 1.044 to 1.34. These are indicative rather than formal checks, since the amplifier applies to sway effects and not uniformly to every utilisation, but the two braces are sway members and for them the figure is close to literal.

One limitation strengthens rather than weakens the finding. Buckling analysis treats tie elements as bars, so both diagonals of each X-brace resist compression in the eigenvalue solution, whereas the design relies on one. The true elastic critical load factor with tension-only bracing is therefore lower than 4.58.

9.2 Frame stability depends on floor diaphragm action

The buckling result proved highly sensitive to how the floors were modelled, and quantifying that sensitivity became a finding in itself.

Floor modelling	Mode 1	Global participation	Modes below 1.0
No floor constraints	0.530	0.35	11
GSA four-corner constraints	0.629	0.07	1
Full-floor diaphragms, four levels	4.576	0.90	0
Full-floor diaphragms, L1 omitted	4.577	0.90	0

Table 9 Sensitivity of the buckling spectrum to floor diaphragm modelling.

Without floor diaphragms the frame is unstable under its own factored load in eleven distinct ways, several of them substantially global. Modelling the floors removes ten of them. The four-corner constraints that GSA generated by default — the same behaviour noted in §4.3 — were almost useless, improving the load factor from 0.53 only to 0.63 while leaving the building twisting 543×10^{-6} radians

under symmetric wind loading. Rebuilding the constraints with complete node lists for each level resolved both.

Critically, removing the Level 1 diaphragm leaves the global mode unchanged to four significant figures. The elastic critical load factor is therefore independent of any assumption about slab restraint at the transfer level, which makes the principal finding robust.

9.3 The truss chord depends on the slab for its effective length

The 0.63 mode present in the corner-constraint model was identified as the truss top chord buckling laterally, unrestrained over its full 24.5 m length. Including the chord panel points in the Level 1 diaphragm removed it entirely.

Level 1 diaphragm	Chord local buckling mode
Absent	1.58 × design load
Present	Above 4.5

Table 10 Truss chord local stability with and without slab restraint.

The design adopts an effective length equal to the panel length, 2750 mm, on the basis that the truss elements are pin jointed. That is valid only if the chord is held laterally at every panel point, which requires the slab. The report does not state this.

The inconsistency is worth setting out carefully, because it is not an error of arithmetic. On page 42, in the bracing section, the same report states that "we can consider the concrete floor slabs of the multi-storey to work as plan bracing components". The slab's in-plane action is therefore invoked when distributing wind to the braced bays, and ignored when establishing the chord effective length. They are the same slab. Modelling confirms the two assumptions are not independent: a rigid diaphragm at Level 1 prevents the top chord shortening axially and therefore prevents the truss bending at all, and the mid-span deflection falls from 30.30 mm to 0.827 mm — a factor of 37 on the same structure under the same load. Each assumption is individually defensible; the combination is not.

9.4 Storey drift

Lateral drift was extracted under the factored wind X combination, with the gravity case subtracted to isolate the wind response.

Storey	Height	Drift	Limit h/300	Utilisation
Ground to L1	7000 mm	9.07 mm	23.3 mm	0.39
L1 to L2	5000 mm	1.67 mm	16.7 mm	0.10
L2 to L3	5000 mm	1.01 mm	16.7 mm	0.06
L3 to Roof	5000 mm	0.63 mm	16.7 mm	0.04

Table 11 Storey drift under the ULS wind X combination. Total roof sway 12.38 mm against H/500 = 44 mm.

All storeys pass comfortably, and since the applied wind pressure already carries the 1.5 factor the serviceability values are a third smaller again. The ground storey dominates for three compounding reasons: it is 7 m rather than 5 m tall, its braces are correspondingly flatter and therefore less efficient horizontally, and it carries the full 270.6 kN of accumulated shear. The original report computes drift nowhere, so this check is entirely additional.

9.5 Corroboration by the code's own drift method

EN 1993-1-1 §5.2.1(4)B permits the elastic critical load factor to be estimated directly from storey drift as αcr = (HEd/VEd)(h/δH). Applying it per storey gives an independent check on the eigenvalue result.

Storey	HEd	VEd	αcr from equation 5.2
Ground to L1	270.6 kN	35,907 kN	5.82
L1 to L2	182.8 kN	25,533 kN	21.4
L2 to L3	109.7 kN	15,159 kN	35.7
L3 to Roof	36.6 kN	4,785 kN	61.0

Table 12 Elastic critical load factor by storey, from the code's drift-based expression.

The ground storey governs at 5.82, against 4.687 from the eigenvalue on the same load basis. The two agree within 24 per cent and differ in the expected direction: equation 5.2 captures storey sway P-delta effects only, whereas the eigenvalue also captures member buckling. The eigenvalue is the lower and therefore the value reported.

9.6 Why the drift check alone would not have found this

Drift passes at a utilisation of 0.39 while the elastic critical load factor sits at 4.58. This appears contradictory and is not. The load factor scales with the ratio HEd/VEd, and this building carries 35.9 MN of gravity load against 270 kN of wind — a ratio of 133 to 1. A frame can be stiff enough that drift is trivial and still be close to instability, because what drives P-delta is the vertical load leaning on a small lateral deflection, not the magnitude of the deflection itself.

A designer checking drift, as most would, would conclude that everything is comfortable. This is precisely why §5.2.1 exists as a separate check, and it is the most transferable lesson in the report.

10 Limitations

The following are stated so that the results can be weighed appropriately. None of them is believed to overturn a finding, but each constrains how far a finding can be pushed.

Members were identified in the analysis model from force magnitudes rather than read from member marks. The identification is consistent with section sizes and load paths throughout, but it has not been independently verified. One anomaly remains unexplained: B4's shear changed by −9.0 per cent against a moment change of −4.3 per cent, where B1, B2 and B3 show shear and moment tracking one another exactly.

Some early output was read in element local rather than global axes, which affects the axis about which C1's 307.5 kNm moment acts and therefore whether the interaction utilisation is 1.044 or 1.311. Plastic section moduli used in that screening were taken from memory rather than verified against published section tables, and the full §6.3.3 interaction with interaction factors from Annex B has not been carried out.

Buckling analysis treats tie elements as bars, biasing the elastic critical load factor upward as discussed in §9.1. P-delta and modal analyses were not run, so the load factor rests on two independent methods rather than three.

Two model states were used: gravity and deflection results come from the model without floor diaphragms, and stability results from the model with them. The sensitivity test in §9.2 demonstrates that the load factor is unaffected by this choice; the deflection results are not, and the model without diaphragms is the appropriate one for them.

11 Conclusions

The original design is safe in the sense that matters most: every discrepancy found in the loading assumptions was conservative. Beam self-weight was over-assumed by a factor of two to six, truss self-weight by 39 per cent, and the resulting design actions are between three and fourteen per cent higher than the structure actually attracts. Material has been spent, not risked.

Three findings are not conservative, and they share a common character — each concerns something the hand method assumes away rather than something it calculates incorrectly.

The transfer truss deflects 30.3 mm, and the floors it supports inherit that displacement, reaching nearly 50 mm of absolute movement under live load. Every member passes its own check. Member-by-member checking cannot detect this.

The truss chord's effective length, and with it the entire truss design, depends on lateral restraint from the floor slab. The design relies on that restraint without stating it, while in a different section it invokes the same slab explicitly for wind distribution. The two assumptions are not independent and the document treats them as though they were.

Most significantly, the elastic critical load factor is 4.58, placing the frame in the range where EN 1993-1-1 requires second-order effects. The design used first-order analysis with buckling length equal to storey height. Amplifying the sway effects takes both braces past unity. A designer checking storey drift would have found nothing wrong.

Returning to the question the project set out to answer: the handoff from BIM to analysis broke the data silently and twice, and it broke the idealisation in a way that could not be papered over — the design's own assumption of pinned connections produces a model that will not solve. What the finite element model then offered in return was not greater precision on the quantities the hand calculation already produced. It was access to four questions the hand method cannot ask at all. That asymmetry, rather than any individual number in this report, is the result worth carrying forward.

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